

Boolean Minimization

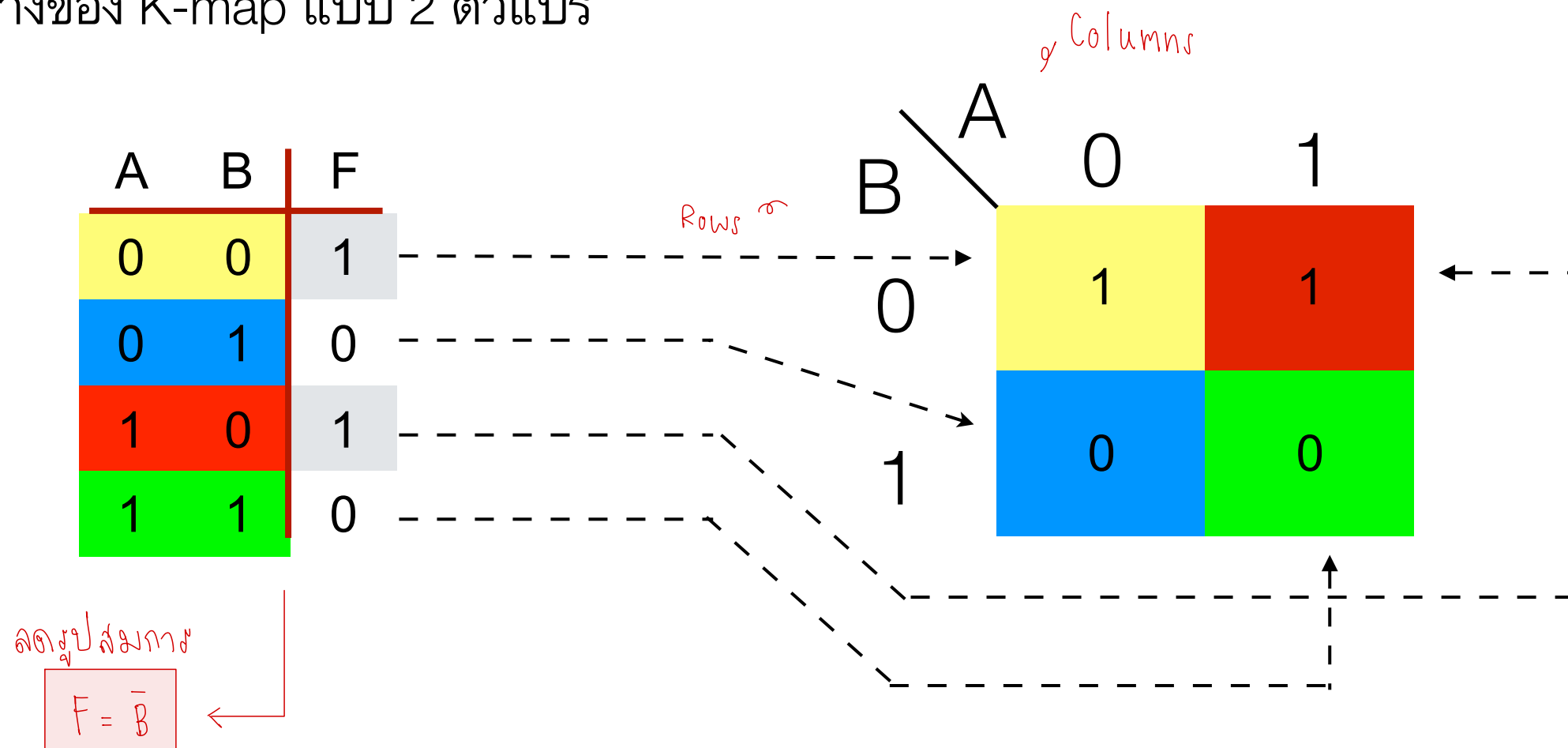
Lecture 4

Outline

- Karnaugh-map
- Circuit design

Karnaugh Map

- Karnaugh Map (K-map) อาศัยการจัดเรียงตารางค่าความจริง ในรูปแบบใหม่ ซึ่งทำให้สามารถลดรูปโดยอาศัยคุณสมบัติการคอมพลิเมนต์ได้ง่ายขึ้น
- ตัวอย่างของ K-map แบบ 2 ตัวแปร



Examples of K-Map

Rows *C* Columns *AB*

	00	01	11	10
0				
1				

ABC

3-variable K-map

CD *AB*

	00	01	11	10
00				
01				
11				
10				

ABCD

4-variable K-map

Using K-Map

- หาช่องใน K-map ที่มีลอจิก 1 ติดกัน
- จำนวนช่องที่ใช้ลดรูปต้องมีขนาด 2, 4, 8, ... เท่านั้น
- ตัวอย่าง

A	B	F
0	0	1
0	1	0
1	0	1
1	1	0

		A	
		0	1
B	0	1	1
	1	0	0

$$F = \overline{B}$$

Using K-Map (2)

- ตัวอย่าง

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

AB \ C	00	01	11	10
0	0	0	1	1
1	0	1	1	1

Output จะเป็น 1 เมื่อ $A=1$ หรือ $B=1$ และ $C=1$

$$F = A + BC$$

SoP

Example

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

		AB			
		00	01	11	10
C	0	0	0	1	0
	1	0	1	1	1

$$F = AB + BC + AC$$

Example (2)

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	1

		AB			
		00	01	11	10
C	0	1	0	0	1
	1	0	0	1	1

$$F = AC + \bar{B}\bar{C}$$

Example (3)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	0

		AB			
	CD	00	01	11	10
00		1	0	0	1
01		1	1	1	1
11		1	0	0	0
10		0	0	0	0

$$F = \bar{C}D + \bar{B}\bar{C} + \bar{A}\bar{B}C$$

Example (4)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	0
0	0	1	0	1
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

		AB			
CD		00	01	11	10
	00	1	0	0	1
01	01	0	1	0	0
11	11	1	1	1	1
10	10	1	1	1	1

$$F = C + \bar{B}\bar{D} + \bar{A}BD$$

Don't Cares

- การใช้ประโยชน์จาก don't cares
- ตัวอย่าง

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	x
1	1	1	x

don't cares

C \ AB				
	00	01	11	10
0	0	0	x	1
1	0	1	x	0

$$F = BC + A\overline{C}$$

Example (5)

A	B	C	F
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	x
1	0	0	x
1	0	1	1
1	1	0	0
1	1	1	1

		AB			
		00	01	11	10
C	0	1	0	0	X
	1	0	X	1	1

$$F = AC + \bar{B}\bar{C}$$

Example (6)

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	x
0	1	0	1	1
0	1	1	0	1
0	1	1	1	1
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	x
1	1	0	1	x
1	1	1	0	x
1	1	1	1	x

AB

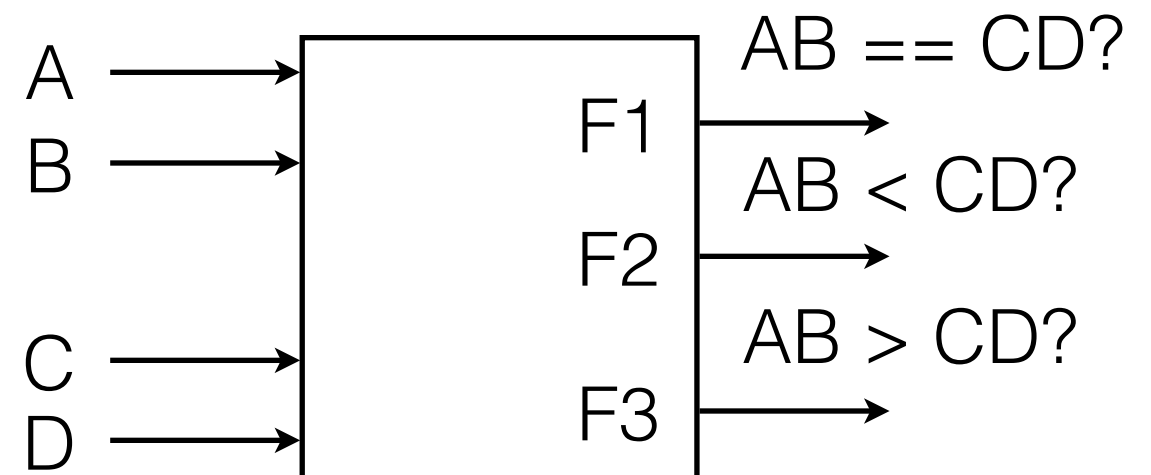
CD

	00	01	11	10
00	1	X	X	1
01	1	1	X	0
11	0	1	X	1
10	1	1	X	0

$$F = \bar{A}\bar{C} + \bar{C}\bar{D} + \bar{A}\bar{D} + B + BCD$$

Circuit Design

- ตัวอย่าง: ออกแบบ Two-bit Comparator
- Input Spec:
 - 2-bit input จำนวน 2 ตัว (AB, CD)
- Output Spec:
 - F1 เช็คว่า $AB == CD$
 - F2 เช็คว่า $AB < CD$
 - F3 เช็คว่า $AB > CD$



Two-bit Comparator

AB == CD?

A	B	C	D	F1	F2	F3
0	0	0	0	1		
0	0	0	1	0		
0	0	1	0	0		
0	0	1	1	0		
0	1	0	0	0		
0	1	0	1	1		
0	1	1	0	0		
0	1	1	1	0		
1	0	0	0	0		
1	0	0	1	0		
1	0	1	0	1		
1	0	1	1	0		
1	1	0	0	0		
1	1	0	1	0		
1	1	1	0	0		
1	1	1	1	1		

CD \ AB	00	01	11	10
00	1	0	0	0
01	0	1	0	0
11	0	0	1	0
10	0	0	0	1

$$F_1 = \bar{A}\bar{B}\bar{C}\bar{D} + \bar{A}\bar{B}\bar{C}D + \bar{A}B\bar{C}\bar{D} + A\bar{B}\bar{C}\bar{D}$$

Two-bit Comparator (2)

AB < CD?

A	B	C	D	F1	F2	F3
0	0	0	0		0	
0	0	0	1		1	
0	0	1	0		1	
0	0	1	1		1	
0	1	0	0		0	
0	1	0	1		0	
0	1	1	0		1	
0	1	1	1		1	
1	0	0	0		0	
1	0	0	1		0	
1	0	1	0		0	
1	0	1	1		1	
1	1	0	0		0	
1	1	0	1		0	
1	1	1	0		0	
1	1	1	1		0	

CD \ AB	00	01	11	10
00	0	0	0	0
01	1	0	0	0
11	1	1	0	1
10	1	1	0	0

$$F_2 = \bar{A}C + \bar{A}\bar{B}D + \bar{B}CD$$

Two-bit Comparator (3)

A	B	C	D	F1	F2	F3
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

→ AB > CD?

CD \ AB	00	01	11	10
00				
01				
11				
10				

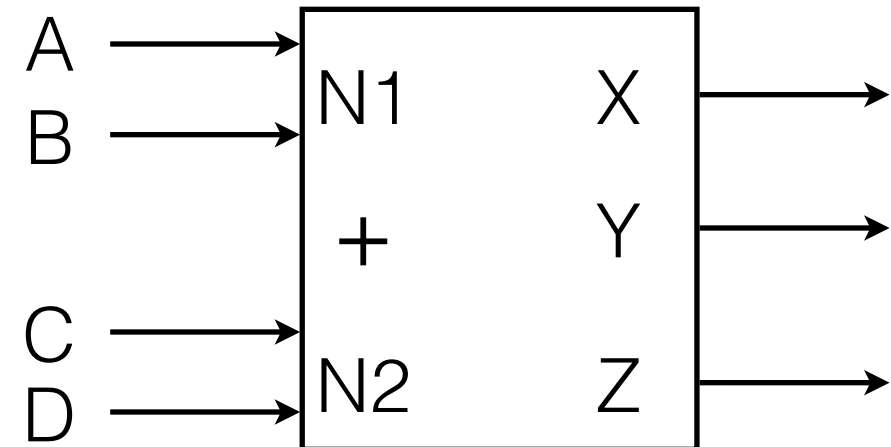
$$F_3 =$$

Two-Bit Comparator (4)

- Schematic Diagram

Two-Bit Binary Addder

- ตัวอย่าง ออกแบบ Two-bit Binary Addder
- Input Spec
 - 2-bit input จำนวน 2 ตัว (AB, CD)
- Output Spec
 - 3-bit binary (XYZ)



Two-bit Binary Adder (2)

A	B	C	D	X	Y	Z
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

CD \ AB	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	1

$$X = AC + BCD + ABD$$

Two-bit Binary Adder (3)

A	B	C	D	X	Y	Z
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

		AB			
CD		00	01	11	10
	00				
	01				
	11				
	10				

$Y =$

Two-bit Binary Adder (4)

A	B	C	D	X	Y	Z
0	0	0	0			
0	0	0	1			
0	0	1	0			
0	0	1	1			
0	1	0	0			
0	1	0	1			
0	1	1	0			
0	1	1	1			
1	0	0	0			
1	0	0	1			
1	0	1	0			
1	0	1	1			
1	1	0	0			
1	1	0	1			
1	1	1	0			
1	1	1	1			

CD \ AB	00	01	11	10
00				
01				
11				
10				

$Z =$

5-Variable K-map

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

ใช้ A ลงใน 2 ตาราง
ไม่เชื่อมกัน

$A = 0$

$A = 1$

		BC			
		00	01	11	10
DE	00	0	1	0	0
	01	1	1	1	1
	11	0	0	1	1
	10	1	1	1	1

		BC			
		00	01	11	10
DE	00	0	0	0	0
	01	1	1	1	1
	11	0	0	1	1
	10	0	0	0	1

$$F = BE + \bar{D}E + \bar{A}DE + B\bar{C}D + \bar{A}\bar{B}\bar{C}\bar{D}$$

5-Variable K-map (2)

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

$A = 0$

DE \ BC	00	01	11	10
00	1	x	x	1
01	0	x	1	0
11	0	1	x	1
10	0	1	x	1

$A = 1$

DE \ BC	00	01	11	10
00	1	1	x	1
01	0	1	1	1
11	0	1	1	1
10	1	1	1	1

$$G = C + AB + \bar{D}\bar{E} + BD + A\bar{E}$$

5-Variable K-map (2)

A	B	C	D	E	F	G
0	0	0	0	0	0	1
0	0	0	0	1	1	0
0	0	0	1	0	1	0
0	0	0	1	1	0	0
0	0	1	0	0	1	x
0	0	1	0	1	1	x
0	0	1	1	0	1	1
0	0	1	1	1	0	1
0	1	0	0	0	0	1
0	1	0	0	1	1	0
0	1	0	1	0	1	1
0	1	0	1	1	1	1
0	1	1	0	0	0	x
0	1	1	0	1	1	1
0	1	1	1	0	1	x
0	1	1	1	1	1	x

A	B	C	D	E	F	G
1	0	0	0	0	0	1
1	0	0	0	1	1	0
1	0	0	1	0	0	0
1	0	0	1	1	0	1
1	0	1	0	0	0	1
1	0	1	0	1	1	1
1	0	1	1	0	0	1
1	0	1	1	1	0	1
1	1	0	0	0	0	1
1	1	0	0	1	1	1
1	1	0	1	0	1	1
1	1	0	1	1	1	1
1	1	1	0	0	0	x
1	1	1	0	1	1	1
1	1	1	1	0	0	1
1	1	1	1	1	1	1

$A = 0$

		BC			
		00	01	11	10
DE	00	1	x	x	1
	01	0	x	1	0
	11	0	1	x	1
	10	0	1	x	1

$A = 1$

		BC			
		00	01	11	10
DE	00	1	1	x	1
	01	0	1	1	1
	11	0	1	1	1
	10	1	1	1	1